

Seasonal Variations in Carbon Budget in Water Column off Princess Astrid Coast, Antarctica

V.K. DHARGALKAR, S.G. PRABHU MATONDKAR and X.N. VERLENCAR

National Institute of Oceanography
Dona Paula, Goa-403004

Abstract

Phytoplankton, zooplankton and particulate matter were studied round the year at a fixed station off Lazarev, Princess Astrid Coast, Antarctica, to assess the impact of prolonged winter on carbon flux. Summer-Winter phytoplankton total counts did not vary much and ranged from 2.8 to 22.8 no. $\times 10^{-4} l^{-1}$. Chlorophyll *a* showed considerable decrease in winter and varied from 0.035 to 0.073 mg m^{-3} and in summer from 0.09 to 3.68 mg m^{-3} . Zooplankton population was high during summer (600 to 2255 no. $100 m^{-3}$) and low in winter (437 to 1184 no. $100 m^{-3}$).

Particulate carbon values ranged from 46 to 83 $\mu g l^{-1}$ during summer and from 22.2 to 63.8 $\mu g l^{-1}$ in winter. Particulate nitrogen showed little variation during summer and winter (5 - 10.15 $\mu g l^{-1}$). Winter effect was particularly seen on the biochemical composition of particulate material when P-CHO and particulate lipid were high during winter and low in summer.

An attempt has been made to evaluate the role of dominant species of phytoplankton and zooplankton in overall composition in POC/PON and C:N of particulate matter round the year in surface and water column upto 100 m water depth.

Introduction

One of the objectives of a biological study is to determine the production and fate of organisms and their byproducts in environment. In Antarctic marine environment, algal matter grows during favourable summer (3-4 months) and organic matter produced act as food for invertebrates for rest of the period. The particulate organic matter is an important component of food available for krill in the Southern Ocean (El-Sayed, 1988). Part of this material sink to bottom supporting benthos in Antarctic waters (Dayton, 1990).

A sizeable amount of data is available on biochemical composition of particulate matter during summer and winter in Antarctic coastal waters (Dhargalkar and Bhosle, 1989 and Dhargalkar and Verlencar, 1992). The attempt is

made here to estimate elemental composition of particulate matter during winter and its role in Antarctic food cycle particularly krill as key organism.

Material and Methods

One station was established in pack ice zone and sampled during winter (May-October) of 1993. The ice was drilled and water was collected using 1 litre Kahlisco bore-hole water sampler with thermometer installed in it. A closing type of net was used for zooplankton sampling. Water sampled at various depths (0-100 m) was used for chl. *a*, phytoplankton counts and composition of particulate matter. Particulate matter retained on GF/F was used for measurement of Total Suspended Matter (TSM), Particulate Organic Carbon (POC), Particulate Organic Nitrogen (PON), carbohydrate, protein and lipids. Calorific content of particulate matter was calculated based on biochemical composition (fractionwise) and total.

Carbon and nitrogen in particulate matter was estimated using PE-2400 CHN analyser and biochemical composition was proximated as described earlier (Dhargalkar and Bhosle, 1989).

Results and Discussions

Data presented here are based on winter collection (under sea ice). Average estimates derived from 5 depth (0 - 100 m) are presented in Tables 1 - 7. Since there is no other work from winter season in Antarctic Ocean on this line the reports of summer work from other area are compared and discussed.

TSM concentrations ranged between 37.0 mg l^{-1} and 133 mg l^{-1} . Higher values were recorded during mid-winter - July (104.0 mg l^{-1}) and August (105.4 mg l^{-1}). However, end of summer - May (63.4 mg l^{-1}) and beginning of Spring - October values were much low (42.0 mg l^{-1}). Summer TSM values were much lower from this area ($29.41 \pm 20.5 \text{ mg l}^{-1}$) as reported by Verlencar *et al.* (1995). Summer TSM values from other regions like Ross Sea ($< 5 \text{ mg l}^{-1}$) and Bransfield Straight ($< 5 \text{ mg l}^{-1}$) were also low (Bolter and Dawsons, 1982; Leibzeit, 1984 and Fabiano *et al.*, 1993). Since TSM values from winter are much higher, the biochemical and elemental composition of the particulate matter in water column is of relevance.

POC and PON content of POM/TSM is presented in Table 1. POC values were most of the time $< 100 \text{ } \mu\text{g l}^{-1}$. The lowest value of $12 \text{ } \mu\text{g l}^{-1}$ was recorded in September i.e. towards the end of winter season. Similarly PON values were not varying widely, i.e., $1 - 10 \text{ } \mu\text{g l}^{-1}$. The general level of POC during winter was lower than for summer which was always $> 100 \text{ } \mu\text{g l}^{-1}$. Same trend was

Table 1: Average and range of the Total Suspended Matter (TSM), Particulate Organic Carbon (POC), Particulate Organic Nitrogen (PON), in first 100 m water column

Month (Winter)	TSM mg l ⁻¹	POC µg l ⁻¹	PON µg l ⁻¹
May	63.4 (37-75)	62.5 (46-83)	10.15 (7.3-12.1)
June	79.2 (64-101)	54.4 (19-131)	5.06 (1.2-6.83)
July	104.4 (59-133)	63.8 (32-95)	9.72 (3.4-18.9)
Aug.	105.4 (102-108)	45.2 (42-77)	6.82 (1.92-13.3)
Sept.	53.0 (46-59)	22.2 (12.-33)	7.932 (3.59-16.42)
Oct.	42 (38-45)	52.4 (23-98)	8.65 (1.0-16.0)

observed in PON also, whereby values ranged from >1 to $8 \mu\text{g l}^{-1}$. (Smith and Nelson, 1985; Tanoue *et al.*, 1986). POC converted to organic matter accounted for negligible fraction of TSM. This is specific to winter season. In a similar work during summer in Ross Sea, 35 - 40% of TSM was organic in nature (Fabiano *et al.*, 1993).

The nature of organic matter during winter was reflected in C:N ratio, which varied from 2.84 in June to 29.2 in July. Bodungen *et al.*, 1986 reported C:N as <10 in Peninsular region during summer. Summer values of Febiano *et al.*, 1993 are also <10 . However, large seasonal variations in C:N have been observed by Nelson and Smith, 1986 and Treguer *et al.*, 1988 as observed in present investigation.

POM and biochemical composition of TSM is presented in Tables 2 to 5. P-CHO was $23 - 138 \mu\text{g l}^{-1}$, P-protein was $24 - 200 \mu\text{g l}^{-1}$ and P-Lipid was $15 - 174 \mu\text{g l}^{-1}$. Variability in carbohydrate, lipid and protein in TSM were primarily associated with the period of high and low standing stock of phytoplankton specially during summer and winter (Table 8). In general, the trends in TSM variations and biochemical components were similar and in agreement with other properties of the TSM such as organic matter, chlorophyll *a*, POC and nitrogen.

Calorific content of POM is presented in Table 5. During peak winter POM was more nutritive ($25.7 \text{ k.cal cm}^{-3}$) as compared to other months ($<10 \text{ k.cal cm}^{-3}$). The food index calculated based on biochemical constituents (Table 3) also indicated nutritive quality of TSM, which was highest during peak winter (112.3 in July) and not during summer. C:N value of TSM during winter was considerably low which satisfies the nutritional requirement of marine animals for protein. The ratio of protein:carbohydrate:lipid (2:3:1) in TSM also meet the food requirement of animals during winter.

Table 2: Average and range of the Particulate Carbohydrate (P-CHO), Particulate Proteins (P-PRT), Particulate Lipids (P-Lip)

Month (Winter)	P-CHO $\mu\text{g l}^{-1}$	P-PRT $\mu\text{g l}^{-1}$	P-LIP $\mu\text{g l}^{-1}$
May	46.5 (23-72)	92 (52-122)	79 (50-120)
June	46.5 (23-72)	54 (40-68)	127 (96-150)
July	98 (62-138)	130 (110-170)	154 (124-174)
Aug.	114 (95-138)	163 (150-200)	116 (90-130)
Sept.	88 (56-110)	29 (24-36)	35 (15-45)
Oct.	83 (74-96)	49 (30-66)	15.5 (18-22)

Table 3: Average and range of carbon in Carbohydrate, Protein, Lipid and Particulate matter ($\mu\text{g l}^{-1}$)

Month	C-PCHO	C-PPRT	C-PLIP	C-POM
May	50.4 (43.2-64.8)	41.1 (24.7-58.0)	5.9 (3.7-9.1)	97.5
June	18.53 (9.2-28.8)	25.7 (19.1-32.3)	9.8 (1.13-11.3)	50.5
July	39.2 (24.8-55.2)	61.8 (52.3-80.9)	11.5 (9.3-13.1)	112.3
Aug.	46.9 (38-55.2)	77.4 (65.7-95.2)	8.73 (6.7-9.8)	99.7
Sept.	35.2 (22.4-46.8)	13.5 (11.4-17.1)	2.63 (1.1-3.39)	51.4
Oct.	33.0 (29.6-38.4)	19.1 (14.2-31.4)	1.36 (1.2-1.66)	56.9

Table 4 : Range and average of different ratios in POM

Month (Winter)	C:N	POC:Chl. <i>a</i>	PPRT:PCHO	NPPr:Chl <i>a</i>
May	(9.25-25.6) 16.45	1276-9600 5382	0.32-1.13 0.796	300-1660 1333
June	1.78-4.4 2.84	21.74-8666 5405	0.75-1.74 1.343	133-1360 853
July	7.1-71.0 29.2	2714-28800 12588	0.79-1.8 1.453	502-7040 3647
Aug.	5.66-8.58 7.22	736-27500 9946	1.0-2.11 1.46	421-11000 4029
Sept.	12-42 25.3	291-1925 957	0.15-0.64 0.356	97-142 121.6
Oct.	12.5-20.1 16.5	7.66-1342 983	0.38-0.89 0.583	76-245 589

Table 5: Caloric content k cal m^{-3} in different fractions (average and range)

Month	Protein	Lipid	Carbohydrate	Total
May	5.06 (2.86-6.71)	4.1 (4.75-11.3)	5.2 (4.4-6.6)	17.6 (15.77-21.33)
June	2.88 (2.2-3.74)	12.1 (9.12-14.25)	1.88 (0.94-2.9)	16.8 (12.2-20.07)
July	7.15 (6.05-9.35)	14.7 (11.78-16.53)	11.46 (2.5-5.6)	25.7 (20.38-29.73)
Aug.	8.61 (7.59-10.0)	11.0 (8.55-12.55)	4.7 (3.8-5.6)	24.3 (21.74-25.8)
Sept.	1.24 (1.32-1.98)	2.95 (1.42-4.29)	2.6 (2.3-4.8)	8.5 (6.6-10.4)
Oct.	2.6 (1.65-3.63)	1.77 (1.52-2.1)	3.4 (3.1-3.9)	7.7 (6.85-8.5)

Table 6: Zooplankton communities (No./100 m^3) collected from Antarctic fixed station

Taxon	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.
Foraminifera	-	20	20	60	50	125	50	250	100
Radiolaria	180	-	100	60	83	375	50	250	145
Anthozoa	-	15	-	-	-	-	-	-	50
Hydrozoa	-	-	-	-	-	50	-	-	-
Siphonophora	125	50	-	20	-	-	-	-	50
Polychaeta	-	-	20	-	-	-	-	-	-
Amphipoda	-	-	-	-	-	-	20	-	60
Copepoda	250	280	70	395	146	250	125	375	480
Decapoda	-	-	-	20	-	-	-	-	-
Crustacean Nauplii	125	125	50	146	125	83	60	625	600
Euphausiae larvae	125	210	78	313	80	200	170	750	550
Eggs	375	-	100	170	20	-	125	-	200
Ophiopluteus larvae	-	-	-	-	-	70	-	-	20
Total	1140	440	438	1184	504	1153	600	2250	2255

Table 7: Phytoplankton species No. $\times 10^{-4} \text{ l}^{-1}$ (Surface only)

No.	Name of the species	May	June	July	Aug.	Sept.	Oct.
1.	<i>Nitzschia</i> sp.	0.8	-	0.4	0.4	5.2	1.6
2.	<i>Fragilaria</i> sp.	3.6	-	-	-	-	-
3.	<i>F. Cylindrus</i>	2.4	5.6	0.8	1.2	3.6	7.2
4.	<i>F. Striatula</i>	2.4	4.0	0.8	0.4	-	-
5.	<i>Melosira sulcata</i>	4.4	-	-	-	-	-
6.	<i>Navicula</i> sp.	2.0	0.8	0.4	0.4	0.8	-
7.	<i>Rhizosolenia</i> sp.	0.4	-	-	-	-	0.4
8.	<i>Corethron</i> sp.	0.4	0.4	-	0.4	-	-
9.	<i>Coscinodiscus</i> sp.	1.2	2.0	-	0.8	0.8	0.8
10.	<i>C. radiatus</i>	-	0.4	-	-	-	-
11.	<i>C. excentrius</i>	-	1.2	-	-	-	-
12.	<i>Chaetoceros</i> sp.	2.4	0.8	-	-	-	0.8
13.	<i>Eucampia</i> sp.	1.2	1.2	-	-	0.8	0.8
14.	<i>Dactylisolen antarcticus</i>	0.8	1.2	-	-	-	-
15.	<i>Thalassiosira</i> sp.	-	-	-	-	-	0.8
16.	<i>Asteromphalus heptactes</i>	-	0.4	0.4	-	-	-
17.	<i>Pinnularia</i> sp.	0.8	0.4	-	-	-	-
18.	<i>Dictyocha speculum</i>	1.2	0.4	-	-	-	-
19.	<i>Biddulphia</i> sp.	-	0.4	-	0.4	-	0.4
20.	<i>Amphora</i> sp.	-	-	-	0.4	-	1.2
21.	Unidentified sp.	0.4	-	-	-	-	-
Total		22.8	19.2	2.8	4.0	11.2	14.0

Table 8: Seasonal variations in the important component of marine ecosystem during 1992-1993 (Surface water)

Parameter	Summer *			Winter		
	Min.	Max.	Mean	Min.	Max.	Mean
TSM (mg l^{-1})	10.5	83.8	28.41	38.0	133.0	75.1
Chl.a (mg m^{-3})	0.09	3.68	1.02	0.03	0.073	0.061
POC ($\mu\text{g l}^{-1}$)	300	1800	450	12	131	49
Zooplankton (No. 100 m^{-3})	10161	150000	50000	438	1184	639
Phytoplankton (Cells $\times 10^{-4} \text{ l}^{-1}$)	0.11	4.64	1.65	2.8	11.2	9.6

* Verlencar et al., 1995.

The results are summarized in Table 8. Total suspended matter (TSM) was more in winter and than in summer. The chlorophyll *a*, and carbon as input components of TSM were more in summer and less in winter. During winter the zooplankton counts were comparatively low, however, phytoplankton counts were equally high during winter (Table 8). This indicated the importance of microheterotrophs during winter, making TSM nutritive.

The quality and quantity of phytoplankton was similar during summer (Verlencar, *et al.*, 1995) and winter (Table 7). Around 20 taxa were prevailing in study area. However, zooplankton growth was retarded during winter reducing the counts of zooplankton from 1 lakh to 1 thousand in 100 m^{-3} of water. The data indicated that the TSM was characteristically nutritious during winter. These results suggest that the gross analysis of the TSM or measurement of a single chemical variable cannot describe the nutritive value of TSM fully. The taxonomic analysis of phytoplankton and microheterotrophs is required for this purpose. Thus, during winter in addition to phyto and zooplankton the role of microheterotrophs need to be considered as a part of biological resource study in Antarctic waters.

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